BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Effectiveness of robotic-assisted therapy for upper extremity in children and adolescents with cerebral palsy: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-045051
Article Type:	Protocol
Date Submitted by the Author:	21-Sep-2020
Complete List of Authors:	Sung-U, Sasithorn; Kobe University, Graduate School of Health Sciences, Department of Rehabilitation Science; Chiang Mai University, Faculty of Associated Medical Sciences, Department of Occupational Therapy Nisa, Badur; Kobe University, Graduate School of Health Sciences, Department of Rehabilitation Science Yotsumoto, Kayano; Kobe University, Graduate School of Health Sciences, Department of Rehabilitation Science Tanemura, Rumi; Kobe University, Graduate School of Health Sciences, Department of Rehabilitation Science
Keywords:	REHABILITATION MEDICINE, Developmental neurology & neurodisability < PAEDIATRICS, Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Full Title: Effectiveness of robotic-assisted therapy for upper extremity in children and adolescents with cerebral palsy: a systematic review protocol

Authors:

Sasithorn Sung-U[1,2], Badur Un Nisa[1], Kayano Yotsumoto[1], Rumi Tanemura[1]

[1] Department of Rehabilitation Science, Graduate School of Health Sciences, Kobe University, Hyogo, Japan

[2] Department of Occupational Therapy, Faculty of Associated Medical Sciences, Chiang Mai University, Chiang Mai, Thailand

Corresponding author:

Sasithorn Sung-U

Department of Rehabilitation Science, Graduate School of Health Sciences, Kobe University, Hyogo, Japan 7-10-2 Tomogaoka, Suma-ku, Kobe, Hyogo, Japan 654 0142

Tel: +8190 3926 9933 Fax: +8178 796 4509

Email: sasithornsungu@gmail.com

Co-authors:

Badur Un Nisa

Email: badurunnisa@yahoo.com

Kayano Yotsumoto

Email: yotumoto@kobe-u.ac.jp

Rumi Tanemura

Email: rumtan@people.kobe-u.ac.jp

For peer review only

Word count: 2,239 (excluding title page, abstract, references, figures and tables)

ABSTRACT

Introduction The application of advanced technologies in pediatric rehabilitation to improve performance and enhance everyday functioning shows considerable promise. The aims of the present systematic review are to investigate the effectiveness of robotic-assisted therapy for upper extremity in children and adolescents with cerebral palsy and to extend the scope of intervention from empirical evidence.

Methods and analysis Multiple databases, including MEDLINE (Ovid), PubMed, CINAHL, Scopus, Web of Science, Cochrane Library, and IEEE Xplore, will be comprehensively searched for relevant randomized controlled trials and non-randomized studies. The grey literature will be accessed on the ProQuest Dissertations & Theses Global database, and a hand search from reference lists of previous articles will be performed. The papers written in English language will be considered, with no limitation on publication date. Two independent reviewers will identify eligible studies, evaluate the level of evidence (the Oxford Center for Evidence-Based Medicine [OCEBM]), and appraise methodological quality and risk of bias (the Standard quality assessment criteria for evaluating primary research papers from a variety of fields [QualSyst tool]; the Grading of Recommendations Assessment, Development and Evaluation [GRADE]). Data will be appropriately extracted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline. A narrative synthesis will be provided to summarize the results, and a meta-analysis will be conducted if there is sufficient homogeneity across outcomes.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Ethics and dissemination Ethical approval is not required for this study. The findings will be disseminated via a peer-reviewed journal and international conferences.

Strengths and limitations of this study

This systematic review protocol focuses specifically on the evidence of the effectiveness and the suitability of service delivery through robotic-assisted therapy for upper extremity in children and adolescents with cerebral palsy.

Intervention outcomes will be described and classified according to the International Classification of Functioning, Disability, and Health, version for Children and Youth (ICF-CY).

We anticipate that meta-analysis could be challenging due to the methodological heterogeneity of feasibility studies.

INTRODUCTION

With a prevalence of approximately 2-3 per 1000 livebirths,[1 2] cerebral palsy (CP) is one of the most common physical and developmental disabilities in childhood. CP is caused by a non-progressive lesion in the developing brain during the prenatal, perinatal, or postnatal periods, and presents lifelong and heterogeneous neurological conditions.[1 3] An acknowledge consensus definition from 2006, describes CP as "a group of permanent disorders of the development of movement and posture, causing activity limitation".[4]

Early signs of CP usually emerge before the first or second year of life, when infants fail to reach key milestones at the expected age.[1 5] Through childhood, the clinical features of

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

neuromotor impairment become more remarkable and variable in types and severity. The manifestations of CP chiefly include abnormalities of muscle tone and reflexes, impaired selective motor control, lack of coordination, atypical posture and movement patterns, and delayed gross and fine motor development. In addition, CP is often accompanied by comorbidities and secondary problems, such as visual and hearing deficits, intellectual disabilities, epilepsy, contracture, and deformities.[1 4 6 7]

Besides symptoms, CP can be displayed within the International Classification of Functioning, Disability and Health (ICF), and the derived version for Children and Youth (ICF-CY) (World Health Organization [WHO]).[8 9] Several studies have reported that children and adolescents with CP exhibit various impairments in body function and structure, and have difficulties performing a wide range of activities.[10 11] The limitations in everyday functioning are common in mobility, self-care,[10 12 13] play,[14 15] school,[16 17] and physical activities.[18 19] Moreover, children and adolescents with CP engage poorly in social and leisure activities comparing to normally developing peers.[20 21] These limitations and restrictions interrelate to a number of contextual factors, which may facilitate or hinder individuals with CP to be more or less independent living and health outcomes.[22 23]

The management of CP requires a multidisciplinary, holistic, and long-term care approaches, involving medical and pharmacological treatments, rehabilitation interventions and assistive technologies alongside complementary and alternative remedies.[24 25] In the 21st century, advanced rehabilitation technologies are more available to patients with neurological disorders, in particular therapeutic robots.[26] Robotic devices have been widely integrated into clinicians' service in order to mediate and assist interventions to enhance clients' recovery and clinical outcomes.[26-28]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

There is a growing need of robotic-assisted therapy in the pediatric field and care delivery for children and adolescents with CP.[29 30] The implementation of robotics primarily focuses on functional motor performance by providing intensive repetitive practice, sensorimotor integration, and cognitive engagement through goal directed tasks.[31] As in previous studies, the use of robotic devices has been found to improve kinematics,[32-34] range of motion,[35] muscle tone,[36 37] postural control,[38] and functions of upper[33] and lower extremities[39] among individuals with CP.

Nonetheless, while the independence in activities and social participation of children and adolescents with CP is inevitably affected by motor dysfunction,[10 11] eighty percent of those having motor impairments of upper extremity.[40] In general, the acquisition of efficient arm and hand skills for use in daily life is a complex process that not only requires neuromusculoskeletal integrity, but also involves various aspects of a child's capabilities. Therefore, apart from positive symptoms that typically present patterns of spasticity, children and adolescents with CP often have a poor ability to reach, grasp, release, and manipulate objects. Furthermore, they experience difficulty using their upper extremities to perform self-care and other activities.[41-43] These problems present great challenges in management and provision of therapeutic intervention, and even in the development and application of innovative technologies such as robotic devices.

Current robotic devices for upper extremity rehabilitation is founded on two main designs. First, an end-effector is a robotic device in which movements are generated from the contact point in the most distal part of the extremity. Second, as a wearable robotic device, an exoskeleton attaches to multiple joints that can reproduce movements in the corresponding

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

parts of the extremity at the same time.[44 45] Both robotic types have been applied to individuals diagnosed with CP and robots now can be used in combination with virtual and augmented reality during training. This could encourage a child's attention and motivation, and may contribute to better performance.[31 46]

Although the use of robots to improve functioning of upper extremities shows promise, the level of evidence in this research area remains lacking. Most of robotic studies in children and adolescents with CP have relatively small sample size and few randomized controlled trials.[30 46] Thus, its effectiveness among this client group is needed to prove clinically. In the past five years, the authors have found three literature reviews focusing on robotic devices for upper extremity function of individuals with CP, but only one has examined the effect size from studies offered sufficient data. The results have shown that robotic-assisted therapy had a moderate effect on reaching duration, smoothness, and muscle tone, and a small to large effect on standardized clinical assessment.[47]

In another systematic research, a prior study has critically reviewed robotic-assisted therapy for locomotion and manipulation in children with CP. The results also indicated a limited number of robotic systems targeting upper extremity.[48] In an earlier study, the researchers have comprehensively focused on robotic devices and protocols for upper extremity in children with neuromotor disorders.[49] Summarize of these reviews laid understanding of the control schemes and state of the art in the implementation of rehabilitation robots, particularly for children who have upper extremity impairments. Nevertheless, the extent of service in the perspective of clinical intervention is still unclear.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

To fill the research gaps, the present systematic review of robotic-assisted therapy for upper extremity may help to verify if this approach is worthwhile for children and adolescents with CP, and how it may be best delivered. In addition, since some recently published studies have reported the beneficial effects of robotic rehabilitation on day-to-day activities, the outcomes will be described and classified into two domains: (1) body function and structure and (2) activity and participation, following the ICF-CY framework.

OBJECTIVES

The objectives of this study are:

1. To investigate the effectiveness of robotic-assisted therapy for upper extremity in children and adolescents with CP on the outcomes within the domains of the ICF-CY.

2. To gain insight into the intervention of robotic-assisted therapy for upper extremity in children and adolescents with CP.

METHOD

Study design

This protocol was developed in accordance with the recommendation of the Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P)[50] and the guideline of the PRISMA statement.[51 52]

Eligibility criteria

Studies will be considered following the Population, Intervention, Comparator, Outcomes, and Study designs or PICOS approach.[52]

Participants

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

We will include the studies involving children and/or adolescents diagnosed with CP in the age range between 0 to 18 years. There are no limitations on gender, types (e.g. spastic hemiplegia/diplegia/quadriplegia, dyskinetic, ataxic, or mixed types), and severity of CP (e.g. mild/moderate/severe CP; levels of gross motor/manual ability/communication functions).

Intervention

Robotic is defined as "the application of electronic, computerized control systems to mechanical devices designed to perform human function".[53] In the present systematic review protocol, robotic-assisted therapy refers to the incorporation of robotic devices targeting the entire or the specific parts of upper extremity into therapeutic training sessions for children and/or adolescents with CP. Studies will be considered either focused on robotic rehabilitation alone or integrated with other approaches, tasks, or technologies (e.g. play, games, or virtual reality). There are no limitations on settings (e.g. clinical-, school-, or home-based settings) and healthcare professionals who provide services (e.g. occupational therapist, physiotherapist, or rehabilitation nurse).

Comparator

Robotic-assisted therapy should be compared to a control intervention, trails with different arms, or experiments with different conditions (e.g. no treatment, placebo, or treatment as usual).

Outcomes

All changes in the scores of outcome measures with at least two times of assessment (e.g. baseline or pretest and posttest or follow up) will be explored for the effect size and the effectiveness of robotic-assisted intervention. These functioning outcomes will be divided

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

into two domains of the ICF-CY framework: (1) body function and structure and (2) activity and participation.

Study types

We will include all randomized controlled trials (RCTs) (e.g. parallel, cluster, and cross-over designs) and non-randomized studies (NRSs) (e.g. cohort, case-control, and case series studies). We will exclude case reports, qualitative studies, and secondary research (e.g. review, systematic review, and meta-analysis).

Search strategy

The relevant articles with English full text will be conducted from electronic databases, including MEDLINE (Ovid), PubMed, CINAHL, Scopus, Web of Science, Cochrane Library, and IEEE Xplore for identification of evidence without date of publication restrictions. The search terms will be initially developed following the search strategy of MEDLINE by combining medical subject headings (MeSH), free text words, and Boolean operators in titles, abstracts, and keywords (Table 1). These terms and strategy will be appropriately modified for other databases. The grey literature will also be accessed on the ProQuest Dissertations & Theses Global database, and a hand search from reference lists of previous papers will be performed. However, the related systematic reviews will be used to facilitate searching of primary sources, but will not include in the final evidence.

Study selection

Initially, database searching and removing duplicate papers will be processed by one reviewer (SS). In the second step, two reviewers (SS, BUN) will independently screen titles and abstracts of the remaining articles to select potentially relevant studies that meet the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

inclusion and exclusion criteria. In the last step, independent reviewers will read the full texts of selected papers to verify their suitability for final review. Any possible disagreement between two reviewers over the eligibility of particular studies will be solved by discussion with a third reviewer (RT). The reasons for excluding study will be recorded. The selection process will be documented in a PRISMA flow diagram.[52]

Data extraction

The data of final included studies will be extracted by one reviewer (SS) and cross-checked by a second reviewer (BUN). If necessary, a consensus will be reached through the arbitration of a third reviewer (RT). A data extraction form will be created to record study characteristic (i.e. authors, publication time, and design), participant information (i.e. types of CP, age, and sample size), intervention delivery (i.e. target of intervention, types of robotic device, modalities, duration and frequency, and settings), comparator (i.e. control, conventional therapy, and/or other alternative treatments), and outcomes within the ICF-CY domains (i.e. body function and structure, and activity and participation). These items will be schemed following the PRISMA guideline.[52]

Quality of the evidence and risk of bias

The quality of the evidence in each included study will be appraised using the standard quality assessment criteria for evaluating primary research papers from a variety of fields (QualSyst tool) for quantitative studies.[54] This tool is a checklist of 14 question items on methodology and risk of bias in research. Each item is scored from 0 to 2 (i.e. “yes” = 2, “partial” = 1, “no” = 0) with a maximum possible score of 28 points. Items which are not applicable to a particular study design are marked as “n/a” and are excluded from the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

summary score calculation. Furthermore, studies will be ranked for the level of evidence corresponding to the Oxford Center for Evidence-Based Medicine (OCEBM).[55]

Additionally, the Grading of Recommendations Assessment, Development and Evaluation or GRADE approach will be used to evaluate the overall quality of evidence across outcomes. To obtain a procedural rating within this framework, the quality of evidence is deliberated in several aspects, starting with the study design (e.g. RCT or observational studies), followed by five reasons to possibly downgrade (i.e. risk of bias, inconsistency, indirectness, imprecision, and publication bias), and three explanations to possibly upgrade (i.e. magnitude of effect, dose of response, and plausible residual confounding). A body of evidence across outcomes will be specified as high, moderate, low, or very low quality.[56]

The same two independent reviewers (SS, BUN) will perform all critical appraisal processes. Discrepancies in judgments will be identified and resolved through discussion with a third reviewer (RT) where necessary.

Data synthesis and analysis

In the part of narrative synthesis, the findings from data extraction will be summarized and reported in table format. A meta-analysis will be conducted from the results of included studies with adequate data by using the Review Manager (RevMan) [Computer program], versions 5.3, The Nordic Cochrane Center, The Cochrane Collaboration.[57]

Ethics and dissemination

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Since this protocol of systematic review will obtain data from relevant published academic papers without deriving from human participants, ethical approval is not required. The findings will be disseminated via a peer-reviewed journal and international conferences.

Author affiliations

[1] Department of Rehabilitation Science, Graduate School of Health Sciences, Kobe University, Hyogo, Japan

[2] Department of Occupational Therapy, Faculty of Associated Medical Sciences, Chiang Mai University, Chiang Mai, Thailand

Contributors SS and RT initiated the conception and designed the study of systematic review protocol. All authors read, provided feedback, and approved the final manuscript.

Competing interests None declared.

Patient and public involvement No patient and public involved.

Patient consent for publication Not required.

ORCID iDs

Sasithorn Sung-U https://orcid.org/0000-0003-0276-8104

Rumi Tanemura https://orcid.org/0000-0002-3390-9080

REFERENCES

For peer review only

1. Patel DR, Neelakantan M, Pandher K, et al. Cerebral palsy in children: a clinical overview. *Transl Pediatr* 2020;9(Suppl 1):S125-S35.
2. Stavsky M, Mor O, Mastrolia SA, et al. Cerebral palsy-trends in epidemiology and recent development in prenatal mechanisms of disease, treatment, and prevention. *Front Pediatr* 2017;5:21.
3. Michael-Asalu A, Taylor G, Campbell H, et al. Cerebral palsy: diagnosis, epidemiology, genetics, and clinical update. *Adv Pediatr* 2019;66:189–208.
4. Rosenbaum P, Paneth N, Leviton A, et al. A report: the definition and classification of cerebral palsy April 2006. *Dev Med Child Neurol* 2007;109(Suppl):8–14.
5. Novak I, Morgan C, Adde L, et al. Early, accurate diagnosis and early intervention in cerebral palsy: advances in diagnosis and treatment. *JAMA Pediatr* 2017;171:897-907.
6. Gulati S, Sondh, V. Cerebral palsy: an overview. *Indian J Pediatr* 2018;85;1006-16.
7. Yamamoto MS. Cerebral palsy. In: Atchison BJ, Dirette DK, eds. *Conditions in occupational therapy: effect on occupational performance*. 4th ed. Baltimore: Lippincott Williams & Wilkins 2012:9-22.
8. World Health Organization. *International classification of functioning, disability and health: ICF*. Geneva: World Health Organization 2001.
9. World Health Organization. *International classification of functioning, disability and health: children and youth version: ICF-CY*. Geneva: World Health Organization 2007.
10. Chan H, Lau P, Fong K, et al. Neuroimpairment, activity, limitation, and participation restriction among children with cerebral palsy in Hong Kong. *Hong Kong Med J* 2005;11:342-50.
11. dos Santos AN, Pavão SL, de Campos AC, et al. International classification of functioning, disability and health in children with cerebral palsy. *Disabil Rehabil* 2012;34:1053-8.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12. Kim K, Kang JY, Jang D. Relationship between mobility and self-care activity in children with cerebral palsy. *Ann Rehabil Med* 2017;41:266-72.
13. Smits DW, Ketelaar M, Gorter JW, et al. Development of daily activities in school-age children with cerebral palsy. *Res Dev Disabil* 2011;32:222–34.
14. Angelin AC, Sposito AM, Pfeifer LI. Influence of functional mobility and manual function on play in preschool children with cerebral palsy. *Hong Kong J Occup Ther* 2018;31:46-53.
15. Graham N, Mandy A, Clarke C, et al. Play experiences of children with a high level of physical disability. *Am J Occup Ther* 2019;73:7306205010p1-10.
16. Huang C, Tseng M, Chen K, et al. Determinants of school activity performance in children with cerebral palsy: a multidimensional approach using the ICF-CY as a framework. *Res Dev Disabil* 2013;34:4025-33.
17. Schenker R, Coster W, Parush S. Personal assistance, adaptations and participation in students with cerebral palsy mainstreamed in elementary schools. *Disabil Rehabil* 2006;28:1061-9.
18. Koldoff EA, Holtzclaw BA. Physical activity among adolescents with cerebral palsy: an integrative review. *J Pediatr Nurs* 2015;30:e105-e17.
19. Lauruschkus K, Westbom L, Hallström I, et al. Physical activity in a total population of children and adolescents with cerebral palsy. *Res Dev Disabil* 2013;34:157-67.
20. Lindsay S. Child and youth experiences and perspectives of cerebral palsy: a qualitative systematic review. *Child Care Health Dev* 2016;42:153-75.
21. Mei C, Reilly S, Reddihough D, et al. Activities and participation of children with cerebral palsy: parent perspectives. *Disabil Rehabil*, 2015;37:2164-73.
22. Imms C. Children with cerebral palsy participate: a review of the literature. *Disabil Rehabil* 2008;30:1867-84.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

23. Parkes J, McCullough N, Madden A. To what extent do children with cerebral palsy participate in everyday life situations? *Health Soc Care Community* 2010;18:304-15.

24. Solopova IA, Moshonkina TR, Umnov VV, et al. Neurorehabilitation of patients with cerebral palsy. *Fiziol Cheloveka* 2015;41:448–54.

25. Trabacca A, Vespino T, Di Liddo A, et al. Multidisciplinary rehabilitation for patients with cerebral palsy: improving long-term care. *J Multidiscip Healthc* 2016;9:455-62.

26. Krebs HI, Volpe BT. Rehabilitation robotics. *Handb Clin Neurol* 2013;110:283-94.

27. Chua K, Kuah C. Innovating with rehabilitation technology in the real world: promises, potentials, and perspectives. *Am J Phys Med Rehabil* 2017;96(10 Suppl 1):S150-6.

28. Laut J, Porfiri M, Raghavan P. The present and future of robotic technology in rehabilitation. *Curr Phys Med Rehabil Rep* 2016;4:312-9.

29. Miguel Cruz A, Ríos Rincón AM, Rodríguez Dueñas WR, et al. What does the literature say about using robots on children with disabilities?. *Disabil Rehabil Assist Technol* 2017;12:429-40.

30. Reyes F, Niedzwecki C, Gaebler-Spira D. Technological advancements in cerebral palsy rehabilitation. *Phys Med Rehabil Clin N Am* 2020;31:117-29.

31. Fasoli SE, Ladenheim B, Mast J, et al. New horizons for robot-assisted therapy in pediatrics. *Am J Phys Med Rehabil* 2012;*91*(11 Suppl 3):S280–9.

32. Aurich-Schuler T, Grob F, van Hedel H, et al. Can Lokomat therapy with children and adolescents be improved? An adaptive clinical pilot trial comparing guidance force, path control, and FreeD. *J Neuroeng Rehabil* 2017;14:76.

33. Biffi E, Maghini C, Cairo B, et al. Movement velocity and fluidity improve after Armeo®Spring rehabilitation in children affected by acquired and congenital brain diseases: an observational study. *Biomed Res Int* 2018;1537170.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

34. Cimolin V, Germiniasi C, Galli M, et al. Robot-assisted upper limb training for hemiplegic children with cerebral palsy. *J Dev Phys Disabil* 2019;31:89-101.

35. Wood KC, Lathan CE, Kaufman, KR. Feasibility of gestural feedback treatment for upper extremity movement in children with cerebral palsy. *IEEE Trans Neural Syst Rehabil Eng* 2013;21:300-5.

36. Fasoli SE, Fragala-Pinkham M, Hughes R, et al. Robotic therapy and botulinum toxin type A: a novel intervention approach for cerebral palsy. *Am J Phys Med Rehabil* 2008;87:1022-5.

37. Frascarelli F, Masia L, Di Rosa G, et al. The impact of robotic rehabilitation in children with acquired or congenital movement disorders. *Eur J Phys Rehabil Med* 2009;45:135-41.

38. Nikityuk IE, Moshonkina TR, Shcherbakova NA, et al. Effect of locomotor training and functional electrical stimulation on postural function in children with severe cerebral palsy. *Fiziol Cheloveka* 2016;42:37-46.

39. Matsuda M, Iwasaki N, Mataki Y, et al. Robot-assisted training using Hybrid Assistive Limb® for cerebral palsy. *Brain Dev* 2018;40:642-8.

40. Makki D, Doudu J, Nixon M. Prevalence and pattern of upper limb involvement in cerebral palsy. *J Child Orthop* 2014;8:215-9.

41. Eliasson A. Improving the use of hands in daily activities. *Phys Occup Ther Pediatr* 2005;25:37-60.

42. Lemmens RJ, Janssen-Potten YJ, Timmermans AA, et al. Arm hand skilled performance in cerebral palsy: activity preferences and their movement components. *BMC Neurol* 2014;14:52.

43. Ferreira HN, Cirne GN, Pereira SA, et al. Upper extremity motor quality evaluation in children with Cerebral Palsy. *Fisioter mov* 2017;30(Suppl 1):277-84.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

44. Maciejasz P, Eschweiler J, Gerlach-Hahn K, et al. A survey on robotic devices for upper limb rehabilitation. *J Neuroeng Rehabil* 2014;11:3.

45. Molteni F, Gasperini G, Cannaviello G, et al. Exoskeleton and end-effector robots for upper and lower limbs rehabilitation: narrative review. *PM R* 2018;10(9 Suppl 2):S174-88.

46. Meyer-Heim A, van Hedel HJ. Robot-assisted and computer-enhanced therapies for children with cerebral palsy: current state and clinical implementation. *Semin Pediatr Neurol* 2013;20:139-45.

47. Chen Y, Howard AM. Effects of robotic therapy on upper-extremity function in children with cerebral palsy: a systematic review. *Dev Neurorehabil* 2016;19:64-71.

48. Bayón C, Raya R, Lerma SL, et al. Robotic therapies for children with cerebral palsy: a systematic review. *Transl Biomed* 2016;7:44.

49. Falzarano V, Marini F, Morasso P, et al. Devices and protocols for upper limb robot-assisted rehabilitation of children with neuromotor disorders. *Appl. Sci.* 2019;9:2689.

50. Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ* 2015;349:g7647.

51. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med* 2009;6:e1000097.

52. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-Analyses of studies that evaluate health care interventions: explanation and elaboration. *PLoS Med* 2009;6:e1000100.

53. Kwakkel G, Kollen BJ, Krebs HI. Effects of robot-assisted therapy on upper limb recovery after stroke: a systematic review. *Neurorehabil Neural Repair* 2008;22:111-21.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

54. Kmet LM, Lee RC, Cook LS. *Standard quality assessment criteria for evaluating primary research from a variety of fields*. Edmonton: Alberta Heritage Foundation for Medical Research 2004.

55. OCEBM Levels of Evidence Working Group. *The Oxford 2011 levels of evidence: Oxford Centre for Evidence-Based Medicine*. http://www.cebm.net/index.aspx?o=5653 (accessed March 2020).

56. Balshem H, Helfand M, Schunemann HJ, et al. GRADE guidelines: 3. Rating the quality of evidence. *J Clin Epidemiol* 2011;64:401-6.

57. Review Manager (RevMan) [Computer program]. Version 5.3. Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration 2014.

Table 1 Search strategy Medline (Ovid)

#	Search
1	cerebral palsy/
2	(hemipleg* or hemipare* or dipleg* or dipare* or quadripleg* or quadripare* or dyskine* or atheto* or dystoni* or chorea* or ataxi*).mp.
3	child/ or adolescent/
4	(child* or preschool or adolescen* or youth or teen* or student* or pediatric*).mp.
5	robot*.mp.
6	(end effector* or exoskeleton*).mp.
7	(robot* adj2 (assist* OR aid* OR mediate* or facilitate*) adj3 (therap* or interven* or rehab* or training)).mp.
8	upper extremity/
9	(upper extremit* or upper limb* or arm or shoulder or elbow or forearm or wrist or hand or finger).mp.
10	1 or 2
11	3 or 4
12	5 or 6 or 7
13	8 or 9
14	10 and 11 and 12 and 13

/ = medical subject headings (MeSH) terms; * = unlimited truncation; mp = multi-purpose; adj2 = terms within 2 words of each other in any order; adj3 = terms within 3 words of each other in any order

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Research Checklist PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to include in a systematic review protocol[a]

Section and topic	#	Checklist item	Page Reported
ADMINISTRATIVE INFORMATION			
Title			
Identification	1a	Identify the report as a protocol of a systematic review	Page 1, Title and abstract
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number	In the editorial process of PROSPERO, CRD (First submission: August 24, 2020)
Authors			
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1, Header
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 12, Contributors
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	5a	Indicate sources of financial or other support for the review	n/a
Sponsor	5b	Provide name for the review funder and/or sponsor	n/a
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	n/a
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Page 5-7, Introduction
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 7, Objectives
METHOD			
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	Page 7-9, Eligibility criteria Page 9, Search strategy
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	Page 9, Search strategy
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Page 9, Search strategy Page 18, Table 1
Study records			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 10, Data extraction
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	Page 9-10, Study selection

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Research Checklist PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to include in a systematic review protocol[a] *(Continued)*

Section and topic	#	Checklist item	Page Reported
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Page 9-10, Study selection and data extraction
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	Page 10, Data extraction
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Page 8, Outcomes
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Page 10-11, Quality of the evidence and risk of bias
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	n/a
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I2, Kendall's tau)	n/a
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	n/a
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Page 11, Data synthesis and analysis
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	n/a
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	Page 11, Quality of the evidence and risk of bias

[a] It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.

From: Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ* 2015;349:g7647.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

BMJ Open

Effectiveness of robotic-assisted therapy for upper extremity function in children and adolescents with cerebral palsy: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-045051.R1
Article Type:	Protocol
Date Submitted by the Author:	15-Dec-2020
Complete List of Authors:	Sung-U, Sasithorn; Kobe University, Graduate School of Health Sciences, Department of Rehabilitation Science; Chiang Mai University, Faculty of Associated Medical Sciences, Department of Occupational Therapy Nisa, Badur; Kobe University, Graduate School of Health Sciences, Department of Rehabilitation Science Yotsumoto, Kayano; Kobe University, Graduate School of Health Sciences, Department of Rehabilitation Science Tanemura, Rumi; Kobe University, Graduate School of Health Sciences, Department of Rehabilitation Science
Primary Subject Heading:	Rehabilitation medicine
Secondary Subject Heading:	Paediatrics
Keywords:	REHABILITATION MEDICINE, Developmental neurology & neurodisability < PAEDIATRICS, Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Full Title: Effectiveness of robotic-assisted therapy for upper extremity function in children and adolescents with cerebral palsy: a systematic review protocol

Authors:

Sasithorn Sung-U[1,2], Badur Un Nisa[1], Kayano Yotsumoto[1], Rumi Tanemura[1]

[1] Department of Rehabilitation Science, Graduate School of Health Sciences, Kobe University, Hyogo, Japan

[2] Department of Occupational Therapy, Faculty of Associated Medical Sciences, Chiang Mai University, Chiang Mai, Thailand

Corresponding author:

Sasithorn Sung-U

Department of Rehabilitation Science, Graduate School of Health Sciences, Kobe University, Hyogo, Japan 7-10-2 Tomogaoka, Suma-ku, Kobe, Hyogo, Japan 654 0142

Tel: +8190 3926 9933 Fax: +8178 796 4509

Email: sasithornsungu@gmail.com

Co-authors:

Badur Un Nisa

Email: badurunnisa@yahoo.com

Kayano Yotsumoto

Email: yotumoto@kobe-u.ac.jp

Rumi Tanemura

Email: rumtan@people.kobe-u.ac.jp

For peer review only

Word count: 2,641 (excluding title page, abstract, references, figures and tables)

ABSTRACT

Introduction The application of advanced technologies in pediatric rehabilitation to improve performance and enhance everyday functioning shows considerable promise. The aims of the present systematic review are to investigate the effectiveness of robotic-assisted therapy for upper extremity function in children and adolescents with cerebral palsy and to extend the scope of intervention from empirical evidence.

Methods and analysis Multiple databases, including MEDLINE (Ovid), PubMed, CINAHL, Scopus, Web of Science, Cochrane Library, and IEEE Xplore, will be comprehensively searched for relevant randomized controlled trials and non-randomized studies. The grey literature will be accessed on the ProQuest Dissertations & Theses Global database, and a hand search from reference lists of previous articles will be performed. The papers written in English language will be considered, with no limitation on publication date. Two independent reviewers will identify eligible studies, evaluate the level of evidence (the Oxford Center for Evidence-Based Medicine [OCEBM]), and appraise methodological quality and risk of bias (the Standard quality assessment criteria for evaluating primary research papers from a variety of fields [QualSyst tool]; the Grading of Recommendations Assessment, Development and Evaluation [GRADE]). Data will be appropriately extracted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline. A narrative synthesis will be provided to summarize the results, and a meta-analysis will be conducted if there is sufficient homogeneity across outcomes.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PROSPERO registration number CRD42020205818.

Ethics and dissemination Ethical approval is not required for this study. The findings will be disseminated via a peer-reviewed journal and international conferences.

Strengths and limitations of this study

This systematic review protocol focuses specifically on the evidence of the effectiveness and the suitability of service delivery through robotic-assisted therapy for upper extremity function in children and adolescents with cerebral palsy.

Intervention outcomes will be described and classified according to the International Classification of Functioning, Disability, and Health, version for Children and Youth (ICF-CY).

We anticipate that meta-analysis could be challenging due to the methodological heterogeneity of feasibility studies.

INTRODUCTION

With a prevalence of approximately 2-3 per 1000 livebirths,[1 2] cerebral palsy (CP) is one of the most common physical and developmental disabilities in childhood. CP is caused by a non-progressive lesion in the developing brain during the prenatal, perinatal, or postnatal periods, and presents lifelong and heterogeneous neurological conditions.[1 3] An acknowledge consensus definition from 2006, describes CP as “a group of permanent disorders of the development of movement and posture, causing activity limitation”.[4]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Early signs of CP usually emerge before the first or second year of life, when infants fail to reach key milestones at the expected age.[1 5] Through childhood, the clinical features of neuromotor impairments become more remarkable and variable in types and severity. The manifestations of CP chiefly include abnormalities of muscle tone and reflexes, impaired selective motor control, lack of coordination, atypical posture and movement patterns, and delayed gross and fine motor development. In addition, CP is often accompanied by comorbidities and secondary problems, such as visual and hearing deficits, intellectual disabilities, epilepsy, contracture, and deformities.[1 4 6 7]

Besides symptoms, CP can be displayed within the International Classification of Functioning, Disability and Health (ICF), and the derived version for Children and Youth (ICF-CY) (World Health Organization [WHO]).[8 9] Several studies have reported that children and adolescents with CP exhibit various impairments in body function and structure, and have difficulties performing a wide range of activities.[10 11] The limitations in everyday functioning are common in mobility, self-care,[10 12 13] play,[14 15] school,[16 17] and physical activities.[18 19] Moreover, children and adolescents with CP engage poorly in social and leisure activities comparing to normally developing peers.[20 21] These limitations and restrictions interrelate to a number of contextual factors, which may facilitate or hinder individuals with CP to be more or less independent living and health outcomes.[22 23]

The management of CP requires a multidisciplinary, holistic, and long-term care approaches, involving medical and pharmacological treatments, rehabilitation interventions and assistive technologies alongside complementary and alternative remedies.[24 25] In the 21st century, advanced rehabilitation technologies are more available to patients with neurological conditions, in particular therapeutic robots.[26] Robotic devices have been widely integrated into

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

clinicians' service in order to mediate and assist interventions to enhance clients' recovery and clinical outcomes resulting from stroke, traumatic brain injuries, multiple sclerosis, spinal cord injury, and other congenital and acquired movement disorders.[26-28]

There is a growing need of robotic-assisted therapy in the pediatric field and care delivery for children and adolescents with CP.[29 30] Based on the applying principles of neural plasticity, motor control, and motor learning, robotic rehabilitation premises in the adaptability of the brain, which may occur spontaneously over time and/or be induced by movement practice after lesion.[31-33] Thus, the implementation of robotics primarily focuses on functional motor performance by providing intensive repetitive training, sensorimotor integration, and cognitive engagement through goal directed tasks to address the underlying symptoms and related problems due to client's neurological conditions.[34-36] From previous studies, the use of robotic devices has been found to improve kinematics,[37-40] range of motion,[41 42] muscle tone,[43 44] postural control,[45 46] and functionality of upper[38 40] and lower extremities[47 48] among individuals with CP.

For upper extremity motor performance, there is evidence that up to eighty percent of CP clients have upper limb involvement and demonstrate limited functions.[49] In general, the acquisition of efficient arm and hand skills for use in daily life is a complex process that not only requires neuromusculoskeletal integrity, but also involves various aspects of a child's capabilities. Therefore, apart from positive symptoms that typically present patterns of spasticity, children and adolescents with CP often have a poor ability to reach, grasp, release, and manipulate objects. Furthermore, they experience difficulty using their upper extremities to perform self-care and other activities.[50-52] These problems present great challenges in

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

management and provision of therapeutic intervention, and even in the development and application of innovative technologies such as robotic devices.

To date, varying interventions are recommended to ameliorate movements and activities in upper extremity for CP clients, for example constraint-induced movement therapy (CIMT), hand-arm bimanual intensive therapy (HABIT), neurodevelopmental therapy (NDT), intramuscular injection of botulinum neurotoxin A (BoNT-A), as well as, robotic-assisted therapy.[36 53-55] Current robotic devices for upper extremity rehabilitation are founded on two main designs. First, an end-effector is a robotic device in which movements are generated from the contact point in the most distal part of the extremity. Second, as a wearable robotic device, an exoskeleton attaches to multiple joints that can reproduce movements in the corresponding parts of the extremity at the same time.[56 57] Both robotic types have been applied to individuals diagnosed with CP and robots now can be used in combination with virtual and augmented reality during training. This could encourage a child's attention and motivation, and may contribute to better performance.[34 58]

Although the use of robots to improve functioning of upper extremities shows promise, the level of evidence in the pediatric group appears less than in adults. Most of robotic studies in children and adolescents with CP have relatively small sample size and few randomized controlled trials.[30 58] Thus, its effectiveness among this client group is needed to prove clinically. In the past five years, the authors have found three literature reviews focusing on robotic devices for upper extremity function of individuals with CP, but only one has examined the effect size from studies offered sufficient data. The results have shown that robotic-assisted therapy had a moderate effect on reaching duration, smoothness, and muscle tone, and a small to large effect on standardized clinical assessment.[59]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In another systematic research, a prior study has critically reviewed robotic-assisted therapy for locomotion and manipulation in children with CP. The results also indicated a limited number of robotic systems targeting upper extremity function.[60] In an earlier study, the researchers have comprehensively focused on robotic devices and protocols for upper extremity function in children with neuromotor disorders.[61] Summaries of these reviews laid understanding of the control schemes and state of the art in the implementation of rehabilitation robots, particularly for children who have upper extremity impairments. However, the extent of service in the perspective of clinical intervention still requires a broader scope to assemble knowledge on robotic-assisted therapy and provide explicit information on complementary and alternative approaches for upper extremity function in people with CP.

To fill the research gaps, the present systematic review of robotic-assisted therapy for upper extremity function may help to verify if this approach is worthwhile for children and adolescents with CP, and how it may be best delivered. In addition, since some recently published studies have reported the beneficial effects of robotic rehabilitation on day-to-day activities, the outcomes will be described and classified into (1) body function and structure and (2) activity and participation, following the ICF-CY framework.

OBJECTIVES

The objectives of this study are:

1. To investigate the effectiveness of robotic-assisted therapy for upper extremity function in children and adolescents with CP on the outcomes within the domains of the ICF-CY.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

2. To gain insight into the intervention of robotic-assisted therapy for upper extremity function in children and adolescents with CP.

METHOD

Study design

This protocol was developed in accordance with the recommendation of the Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P)[62] and the guideline of the PRISMA statement.[63 64]

Eligibility criteria

Studies will be considered following the Population, Intervention, Comparator, Outcomes, and Study designs or PICOS approach.[64]

Participants

We will include the studies involving children and/or adolescents diagnosed with CP in the age range between 0 to 18 years. There are no limitations on gender, types (e.g. spastic hemiplegia/diplegia/quadriplegia, dyskinetic, ataxic, or mixed types), and severity of CP (e.g. mild/moderate/severe CP; levels of gross motor/manual ability/communication functions).

Intervention

Robotic is defined as "the application of electronic, computerized control systems to mechanical devices designed to perform human function".[65] In the present systematic review protocol, robotic-assisted therapy refers to the incorporation of robotic devices targeting the entire or the specific parts of upper extremity into therapeutic training sessions for children and/or adolescents with CP. Studies will be considered either focused on robotic

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

rehabilitation alone or integrated with other approaches, tasks, or technologies (e.g. play, games, or virtual reality). There are no limitations on settings (e.g. clinical-, school-, or home-based settings) and healthcare professionals who provide services (e.g. occupational therapist, physiotherapist, or rehabilitation nurse).

Comparator

Robotic-assisted therapy should be compared to a control intervention, trails with different arms, or experiments with different conditions (e.g. no treatment, placebo, or treatment as usual).

Outcomes

All changes in the scores of outcome measures with at least two times of assessment (e.g. baseline or pretest and posttest or follow up) will be explored for the effect size and the effectiveness of robotic-assisted intervention. These functioning outcomes will be categorized according to the components and domains of ICF-CY framework,[9 66-68] as follows:

(1) Body function and structure will encompass the measures of functioning at the level of the body systems or body parts, particularly neuromusculoskeletal and movement-related functions of upper extremity. For example, the Range of Motion (ROM) Test, the Manual Muscle Test (MMT), the Ashworth Scale (AS) or Modified Ashworth Scale (MAS), the Fugl-Meyer Assessment for Upper Extremity (FMA-UE), and the Three-Dimensional (3-D) Motion Analysis.

(2) Activity and participation will comprise the measures of functioning at the level of an individual in performing tasks or actions and in involving life situations. The domains of

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

outcomes will focus especially on the ability to grasp, release, and manipulate objects and ability to use upper extremity for self-care and other day-to-day activities. For instance, the ABILHAND-Kids, the Assisting Hand Assessment (AHA), the Melbourne Assessment of Upper Limb Function (MAULF), and the Functional Independence Measure for Children (WeeFIM), and the Children's Assessment of Participation and Enjoyment (CAPE).

Study types

We will include all randomized controlled trials (RCTs) (e.g. parallel, cluster, and cross-over designs) and non-randomized studies (NRSs) (e.g. cohort, case-control, and case series studies). We will exclude case reports, qualitative studies, and secondary research (e.g. review, systematic review, and meta-analysis).

Search strategy

The relevant articles with English full text will be conducted from electronic databases, including MEDLINE (Ovid), PubMed, CINAHL, Scopus, Web of Science, Cochrane Library, and IEEE Xplore for identification of evidence without date of publication restrictions. The search terms will be initially developed following the search strategy of MEDLINE by combining medical subject headings (MeSH), free text words, and Boolean operators in titles, abstracts, and keywords (Table 1). These terms and strategy will be appropriately modified for other databases. The grey literature will also be accessed on the ProQuest Dissertations & Theses Global database, and a hand search from reference lists of previous papers will be performed. However, the related systematic reviews will be used to facilitate searching of primary sources, but will not include in the final evidence.

Study selection

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Initially, database searching and removing duplicate papers will be processed by one reviewer (SS). In the second step, two reviewers (SS, BUN) will independently screen titles and abstracts of the remaining articles to select potentially relevant studies that meet the inclusion and exclusion criteria. In the last step, independent reviewers will read the full texts of selected papers to verify their suitability for final review. Any possible disagreement between two reviewers over the eligibility of particular studies will be solved by discussion with a third reviewer (RT). The reasons for excluding study will be recorded. The selection process will be documented in a PRISMA flow diagram.[64]

Data extraction

The data of final included studies will be extracted by one reviewer (SS) and cross-checked by a second reviewer (BUN). If necessary, a consensus will be reached through the arbitration of a third reviewer (RT). A data extraction form will be created to record study characteristic (i.e. authors, publication time, and design), participant information (i.e. types of CP, age, and sample size), intervention delivery (i.e. target of intervention, types of robotic device, modalities, duration and frequency, and settings), comparator (i.e. control, conventional therapy, and/or other alternative treatments), and outcomes within the ICF-CY domains (i.e. body function and structure, and activity and participation). These items will be schemed following the PRISMA guideline.[64]

Quality of the evidence and risk of bias

The quality of the evidence in each included study will be appraised using the standard quality assessment criteria for evaluating primary research papers from a variety of fields (QualSyst tool) for quantitative studies.[69] This tool is a checklist of 14 question items on methodology and risk of bias in research. Each item is scored from 0 to 2 (i.e. “yes” = 2,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

"partial" = 1, "no" = 0) with a maximum possible score of 28 points. Items which are not applicable to a particular study design are marked as "n/a" and are excluded from the summary score calculation. Furthermore, studies will be ranked for the level of evidence corresponding to the Oxford Center for Evidence-Based Medicine (OCEBM).[70]

Additionally, the Grading of Recommendations Assessment, Development and Evaluation or GRADE approach will be used to evaluate the overall quality of evidence across outcomes. To obtain a procedural rating within this framework, the quality of evidence is deliberated in several aspects, starting with the study design (e.g. RCT or observational studies), followed by five reasons to possibly downgrade (i.e. risk of bias, inconsistency, indirectness, imprecision, and publication bias), and three explanations to possibly upgrade (i.e. magnitude of effect, dose of response, and plausible residual confounding). A body of evidence across outcomes will be specified as high, moderate, low, or very low quality.[71]

The same two independent reviewers (SS, BUN) will perform all critical appraisal processes. Discrepancies in judgments will be identified and resolved through discussion with a third reviewer (RT) where necessary.

Data synthesis and analysis

In the part of narrative synthesis, the results from data extraction will be reported in text and table format to summarize the characteristics and findings of the included studies. For a description of outcome measures within the ICF-CY framework, outcome data of each study will be addressed in accordance with the body function and structure and/or the activity and participation domains.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

A meta-analysis will be conducted if there are sufficient data from the included studies. The risk ratio (RR) and odds ratio (OR) will be calculated for dichotomous outcome data. For continuous outcome data, the mean difference (MD) will be used to compare data measured on the same scales while the standard mean difference (SMD) will be used to summarize data measured on the different scales. Heterogeneity will be evaluated by using the inconsistency index (I^2) statistic to describe the percentages of total variation across studies ($I^2 \leq 50\%$ = low heterogeneity; $I^2 > 50\%$ = moderate to high heterogeneity). Where appropriate for pooling effect sizes, a fixed effects model will be conducted when heterogeneity is low, and a random effects model will be performed when heterogeneity is moderate to high.[72 73] If unable to pool results, the data will be reported descriptively according to the classification of outcome measures. The quantitative synthesis will be analyzed by using the Review Manager (RevMan) [Computer program], versions 5.3, The Nordic Cochrane Center, The Cochrane Collaboration.[74]

Ethics and dissemination

Since this protocol of systematic review will obtain data from relevant published academic papers without deriving from human participants, ethical approval is not required. The findings will be disseminated via a peer-reviewed journal and international conferences.

Contributors SS coordinated the study. SS and RT initiated the conception and designed the study of systematic review protocol. SS and BUN developed the search strategy. RT and KY provided instruction on important intellectual contributions. All authors read, provided feedback, and approved the final version of the manuscript.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Funding This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Competing interests None declared.

Patient and public involvement No patient and public involved.

Patient consent for publication Not required.

ORCID iDs

Sasithorn Sung-U https://orcid.org/0000-0003-0276-8104

Rumi Tanemura https://orcid.org/0000-0002-3390-9080

REFERENCES

1. Patel DR, Neelakantan M, Pandher K, et al. Cerebral palsy in children: a clinical overview. *Transl Pediatr* 2020;9(Suppl 1):S125-S35.
2. Stavsky M, Mor O, Mastrolia SA, et al. Cerebral palsy-trends in epidemiology and recent development in prenatal mechanisms of disease, treatment, and prevention. *Front Pediatr* 2017;5:21.
3. Michael-Asalu A, Taylor G, Campbell H, et al. Cerebral palsy: diagnosis, epidemiology, genetics, and clinical update. *Adv Pediatr* 2019;66:189–208.
4. Rosenbaum P, Paneth N, Leviton A, et al. A report: the definition and classification of cerebral palsy April 2006. *Dev Med Child Neurol* 2007;109(Suppl):8–14.
5. Novak I, Morgan C, Adde L, et al. Early, accurate diagnosis and early intervention in cerebral palsy: advances in diagnosis and treatment. *JAMA Pediatr* 2017;171:897-907.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

6. Gulati S, Sondh, V. Cerebral palsy: an overview. *Indian J Pediatr* 2018;85;1006-16.

7. Yamamoto MS. Cerebral palsy. In: Atchison BJ, Dirette DK, eds. *Conditions in occupational therapy: effect on occupational performance*. 4th ed. Baltimore: Lippincott Williams & Wilkins 2012:9-22.

8. World Health Organization. *International classification of functioning, disability and health: ICF*. Geneva: World Health Organization 2001.

9. World Health Organization. *International classification of functioning, disability and health: children and youth version: ICF-CY*. Geneva: World Health Organization 2007.

10. Chan H, Lau P, Fong K, et al. Neuroimpairment, activity, limitation, and participation restriction among children with cerebral palsy in Hong Kong. *Hong Kong Med J* 2005;11:342-50.

11. dos Santos AN, Pavão SL, de Campos AC, et al. International classification of functioning, disability and health in children with cerebral palsy. *Disabil Rehabil* 2012;34:1053-8.

12. Kim K, Kang JY, Jang D. Relationship between mobility and self-care activity in children with cerebral palsy. *Ann Rehabil Med* 2017;41:266-72.

13. Smits DW, Ketelaar M, Gorter JW, et al. Development of daily activities in school-age children with cerebral palsy. *Res Dev Disabil* 2011;32:222–34.

14. Angelin AC, Sposito AM, Pfeifer LI. Influence of functional mobility and manual function on play in preschool children with cerebral palsy. *Hong Kong J Occup Ther* 2018;31:46-53.

15. Graham N, Mandy A, Clarke C, et al. Play experiences of children with a high level of physical disability. *Am J Occup Ther* 2019;73:7306205010p1-10.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

16. Huang C, Tseng M, Chen K, et al. Determinants of school activity performance in children with cerebral palsy: a multidimensional approach using the ICF-CY as a framework. *Res Dev Disabil* 2013;34:4025-33.
17. Schenker R, Coster W, Parush S. Personal assistance, adaptations and participation in students with cerebral palsy mainstreamed in elementary schools. *Disabil Rehabil* 2006;28:1061-9.
18. Koldoff EA, Holtzclaw BA. Physical activity among adolescents with cerebral palsy: an integrative review. *J Pediatr Nurs* 2015;30:e105-e17.
19. Lauruschkus K, Westbom L, Hallström I, et al. Physical activity in a total population of children and adolescents with cerebral palsy. *Res Dev Disabil* 2013;34:157-67.
20. Lindsay S. Child and youth experiences and perspectives of cerebral palsy: a qualitative systematic review. *Child Care Health Dev* 2016;42:153-75.
21. Mei C, Reilly S, Reddihough D, et al. Activities and participation of children with cerebral palsy: parent perspectives. *Disabil Rehabil* 2015;37:2164-73.
22. Imms C. Children with cerebral palsy participate: a review of the literature. *Disabil Rehabil* 2008;30:1867-84.
23. Parkes J, McCullough N, Madden A. To what extent do children with cerebral palsy participate in everyday life situations? *Health Soc Care Community* 2010;18:304-15.
24. Solopova IA, Moshonkina TR, Umnov VV, et al. Neurorehabilitation of patients with cerebral palsy. *Fiziol Cheloveka* 2015;41:448–54.
25. Trabacca A, Vespino T, Di Liddo A, et al. Multidisciplinary rehabilitation for patients with cerebral palsy: improving long-term care. *J Multidiscip Healthc* 2016;9:455-62.
26. Krebs HI, Volpe BT. Rehabilitation robotics. *Handb Clin Neurol* 2013;110:283-94.
27. Chua K, Kuah C. Innovating with rehabilitation technology in the real world: promises, potentials, and perspectives. *Am J Phys Med Rehabil* 2017;96(10 Suppl 1):S150-6.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

28. Laut J, Porfiri M, Raghavan P. The present and future of robotic technology in rehabilitation. *Curr Phys Med Rehabil Rep* 2016;4:312-9.

29. Miguel Cruz A, Ríos Rincón AM, Rodríguez Dueñas WR, et al. What does the literature say about using robots on children with disabilities?. *Disabil Rehabil Assist Technol* 2017;12:429-40.

30. Reyes F, Niedzwecki C, Gaebler-Spira D. Technological advancements in cerebral palsy rehabilitation. *Phys Med Rehabil Clin N Am* 2020;31:117-29.

31. Huang VS, Krakauer JW. Robotic neurorehabilitation: a computational motor learning perspective. *J Neuroeng Rehabil* 2009;6:5.

32. Iandolo R, Marini F, Semprini M, et al. Perspectives and challenges in robotic neurorehabilitation. *Appl Sci* 2019;9:3183.

33. Turner DL, Ramos-Murguialday A, Birbaumer N, et al. Neurophysiology of robot-mediated training and therapy: a perspective for future use in clinical populations. *Front Neurol* 2013;4:184.

34. Fasoli SE, Ladenheim B, Mast J, et al. New horizons for robot-assisted therapy in pediatrics. *Am J Phys Med Rehabil* 2012;91(11 Suppl 3):S280–9.

35. Gassert R, Dietz V. Rehabilitation robots for the treatment of sensorimotor deficits: a neurophysiological perspective. *J Neuroeng Rehabil* 2018;15:46.

36. Reid LB, Rose SE, Boyd RN. Rehabilitation and neuroplasticity in children with unilateral cerebral palsy. *Nat Rev Neurol* 2015;11:390-400.

37. Aurich-Schuler T, Grob F, van Hedel H, et al. Can Lokomat therapy with children and adolescents be improved? An adaptive clinical pilot trial comparing guidance force, path control, and FreeD. *J Neuroeng Rehabil* 2017;14:76.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

38. Biffi E, Maghini C, Cairo B, et al. Movement velocity and fluidity improve after Armeo®Spring rehabilitation in children affected by acquired and congenital brain diseases: an observational study. *Biomed Res Int* 2018;1537170.

39. Cimolin V, Germiniasi C, Galli M, et al. Robot-assisted upper limb training for hemiplegic children with cerebral palsy. *J Dev Phys Disabil* 2019;31:89-101.

40. Gilliaux M, Renders A, Dispa D, et al. Upper limb robot-assisted therapy in cerebral palsy: a single-blind randomized controlled trial. *Neurorehabil Neural Repair* 2015;29:183-92.

41. Wood KC, Lathan CE, Kaufman, KR. Feasibility of gestural feedback treatment for upper extremity movement in children with cerebral palsy. *IEEE Trans Neural Syst Rehabil Eng* 2013;21:300-5.

42. Bayón C, Martín-Lorenzo T, Moral-Saiz B, et al. A robot-based gait training therapy for pediatric population with cerebral palsy: goal setting, proposal and preliminary clinical implementation. *J Neuroeng Rehabil* 2018;15:69.

43. Fasoli SE, Fragala-Pinkham M, Hughes R, et al. Robotic therapy and botulinum toxin type A: a novel intervention approach for cerebral palsy. *Am J Phys Med Rehabil* 2008;87:1022-5.

44. Frascarelli F, Masia L, Di Rosa G, et al. The impact of robotic rehabilitation in children with acquired or congenital movement disorders. *Eur J Phys Rehabil Med* 2009;45:135-41.

45. Nikityuk IE, Moshonkina TR, Shcherbakova NA, et al. Effect of locomotor training and functional electrical stimulation on postural function in children with severe cerebral palsy. *Fiziol Cheloveka* 2016;42:37-46.

46. Drużbicki M, Rusek W, Szczepanik M, et al. Assessment of the impact of orthotic gait training on balance in children with cerebral palsy. *Acta Bioeng Biomech* 2010;12:53-8.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

47. Matsuda M, Iwasaki N, Mataki Y, et al. Robot-assisted training using Hybrid Assistive Limb® for cerebral palsy. *Brain Dev* 2018;40:642-8.

48. Wallard L, Dietrich G, Kerlirzin Y, et al. Robotic-assisted gait training improves walking abilities in diplegic children with cerebral palsy. *Eur J Paediatr Neurol* 2017;21:557-64.

49. Makki D, Doudu J, Nixon M. Prevalence and pattern of upper limb involvement in cerebral palsy. *J Child Orthop* 2014;8:215-9.

50. Eliasson A. Improving the use of hands in daily activities. *Phys Occup Ther Pediatr* 2005;25:37-60.

51. Lemmens RJ, Janssen-Potten YJ, Timmermans AA, et al. Arm hand skilled performance in cerebral palsy: activity preferences and their movement components. *BMC Neurol* 2014;14:52.

52. Ferreira HN, Cirne GN, Pereira SA, et al. Upper extremity motor quality evaluation in children with Cerebral Palsy. *Fisioter mov* 2017;30(Suppl 1):277-84.

53. Plasschaert VFP, Vriezekolk JE, Aarts PBM, et al. Interventions to improve upper limb function for children with bilateral cerebral palsy: a systematic review. *Dev Med Child Neurol* 2019;61:899-907.

54. Sakzewski L, Gordon A, Eliasson AC. The state of the evidence for intensive upper limb therapy approaches for children with unilateral cerebral palsy. *J Child Neurol* 2014;29:1077-90.

55. Sakzewski L, Ziviani J, Boyd R. Systematic review and meta-analysis of therapeutic management of upper-limb dysfunction in children with congenital hemiplegia. *Pediatrics* 2009;123:e1111-22.

56. Maciejasz P, Eschweiler J, Gerlach-Hahn K, et al. A survey on robotic devices for upper limb rehabilitation. *J Neuroeng Rehabil* 2014;11:3.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

57. Molteni F, Gasperini G, Cannaviello G, et al. Exoskeleton and end-effector robots for upper and lower limbs rehabilitation: narrative review. *PM R* 2018;10(9 Suppl 2):S174-88.

58. Meyer-Heim A, van Hedel HJ. Robot-assisted and computer-enhanced therapies for children with cerebral palsy: current state and clinical implementation. *Semin Pediatr Neurol* 2013;20:139-45.

59. Chen Y, Howard AM. Effects of robotic therapy on upper-extremity function in children with cerebral palsy: a systematic review. *Dev Neurorehabil* 2016;19:64-71.

60. Bayón C, Raya R, Lerma SL, et al. Robotic therapies for children with cerebral palsy: a systematic review. *Transl Biomed* 2016;7:44.

61. Falzarano V, Marini F, Morasso P, et al. Devices and protocols for upper limb robot-assisted rehabilitation of children with neuromotor disorders. *Appl. Sci.* 2019;9:2689.

62. Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ* 2015;349:g7647.

63. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med* 2009;6:e1000097.

64. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-Analyses of studies that evaluate health care interventions: explanation and elaboration. *PLoS Med* 2009;6:e1000100.

65. Kwakkel G, Kollen BJ, Krebs HI. Effects of robot-assisted therapy on upper limb recovery after stroke: a systematic review. *Neurorehabil Neural Repair* 2008;22:111-21.

66. Haberfehlner H, Goudriaan M, Bonouvrié LA, et al. Instrumented assessment of motor function in dyskinetic cerebral palsy: a systematic review. *J NeuroEngineering Rehabil* 2020;17:39.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

67. Schiariti V, Klassen AF, Cieza A, et al. Comparing contents of outcome measures in cerebral palsy using the International Classification of Functioning (ICF-CY): a systematic review. *Eur J Paediatr Neurol* 2014;18:1-12.

68. Wallen M, Stewart K. Grading and quantification of upper extremity function in children with spasticity. *Semin Plast Surg* 2016;30:5-13.

69. Kmet LM, Lee RC, Cook LS. *Standard quality assessment criteria for evaluating primary research from a variety of fields*. Edmonton: Alberta Heritage Foundation for Medical Research 2004.

70. OCEBM Levels of Evidence Working Group. *The Oxford 2011 levels of evidence: Oxford Centre for Evidence-Based Medicine*. http://www.cebm.net/index.aspx?o=5653 (accessed March 2020).

71. Balshem H, Helfand M, Schunemann HJ, et al. GRADE guidelines: 3. Rating the quality of evidence. *J Clin Epidemiol* 2011;64:401-6.

72. Higgins JPT, Thomas J, Chandler J, et al. *Cochrane Handbook for Systematic Reviews of Interventions*. 2nd ed. Chichester: John Wiley & Sons 2019.

73. Higgins JPT, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557-60.

74. Review Manager (RevMan) [Computer program]. Version 5.3. Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration 2014.

Table 1 Search strategy Medline (Ovid)

#	Search
1	cerebral palsy/
2	(hemipleg* or hemipare* or dipleg* or dipare* or quadripleg* or quadripare* or dyskine* or atheto* or dystoni* or chorea* or ataxi*).mp.
3	child/ or adolescent/
4	(child* or preschool or adolescen* or youth or teen* or student* or pediatric*).mp.
5	robot*.mp.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

6	(end effector* or exoskeleton*).mp.
7	(robot* adj2 (assist* OR aid* OR mediate* or facilitate*) adj3 (therap* or interven* or rehab* or training)).mp.
8	upper extremity/
9	(upper extremit* or upper limb* or arm or shoulder or elbow or forearm or wrist or hand or finger).mp.
10	1 or 2
11	3 or 4
12	5 or 6 or 7
13	8 or 9
14	10 and 11 and 12 and 13

/ = medical subject headings (MeSH) terms; * = unlimited truncation; mp = multi-purpose; adj2 = terms within 2 words of each other in any order; adj3 = terms within 3 words of each other in any order

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Research Checklist PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to include in a systematic review protocol[a]

Section and topic	#	Checklist item	Page reported
ADMINISTRATIVE INFORMATION			
Title			
Identification	1a	Identify the report as a protocol of a systematic review	Page 1, Title and abstract
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number	Page 3, PROSPERO registration number CRD 42020205818
Authors			
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1, Header
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 13, Contributors
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	5a	Indicate sources of financial or other support for the review	n/a
Sponsor	5b	Provide name for the review funder and/or sponsor	n/a
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	n/a
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Page 5-7, Introduction
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 7, Objectives
METHOD			
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	Page 8-10, Eligibility criteria
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	Page 10, Search strategy
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Page 10, Search strategy Page 21, Table 1
Study records			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 11, Data extraction
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	Page 10-11, Study selection

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Research Checklist PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to include in a systematic review protocol[a] *(Continued)*

Section and topic	#	Checklist item	Page reported
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Page 10-11, Study selection and data extraction
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	Page 11, Data extraction
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Page 9-10, Outcomes
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Page 11-12, Quality of the evidence and risk of bias
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	Page 12-13, Data synthesis and analysis
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I2, Kendall's tau)	Page 12-13, Data synthesis and analysis
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	n/a
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Page 12-13, Data synthesis and analysis
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	n/a
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	Page 12, Quality of the evidence and risk of bias

[a] It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.

From: Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ* 2015;349:g7647.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml